

Erik Löffelholz

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PROFILE

Software developer and computational scientist with a Master's in Mathematical Physics. I work mostly in Python and PyTorch, but I am comfortable across the stack: GPU compute on one end, React and Three.js frontends on the other. Most of what I have built is scientific software, including differentiable simulators, numerical solvers, and interactive 3D tools. I also work day to day with agentic coding models and LLM-assisted development.

EDUCATION

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| Oct 2021 - Sep 2025 | M.Sc. Mathematical Physics
Universität Leipzig
Focus: Differential Geometry · PDE Theory · Quantum Field Theory · Numerical Methods |
| Oct 2018 - Mar 2022 | B.Sc. Physics
Universität Leipzig |

PROFESSIONAL EXPERIENCE

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| Apr 2025 - May 2026 | Scientific Software Developer
Max Planck Institute for Mathematics in the Sciences, Leipzig
Started as a student developer and was promoted to Scientist for the last two months. <ul style="list-style-type: none">Built GPU-accelerated differentiable simulation and 3D visualization software (PyTorch, Three.js, WebGL) for mathematical researchImplemented a discrete differential geometry operator library and pipelines for embedding meshes into curved (hyperbolic and spherical) spaces by energy minimizationDesigned a binary WebSocket protocol that connected Python and PyTorch compute to the browser, roughly 10 to 20 times faster than JSON, with CUDA, MPS and CPU backendsCo-authored a research paper on hyperbolic surface mesh embeddings (Bridges Conference 2026, under review) |
| 2025 - Present | Independent Developer & Researcher
Graph ML, Differentiable Physics & Scientific Computing <ul style="list-style-type: none">Built full-stack ML and simulation apps (PyTorch, FastAPI, React and Three.js): a graph ML framework from scratch, a PINN solver for ten PDEs, and browser-based physics demosWork day to day with agentic coding models and LLM-assisted development |
| 2025 - Present | Subject-Matter Expert, STEM & Coding
Outlier.ai, AI Training & Model Evaluation <ul style="list-style-type: none">Wrote and reviewed STEM, mathematics and physics tasks for large-language-model trainingRated and ranked model outputs (RLHF) and did coding tasks in Python, C++ and JavaScriptContributed German (de-DE) prompt and audio-prompt tasks |
| Mar 2024 - Feb 2025 | Working Student, Mathematics Editorial
Ernst Klett Verlag GmbH, Leipzig <ul style="list-style-type: none">Reviewed mathematics content for school textbooksChecked technical correctness and clarity across several titles |

TECHNICAL SKILLS

Languages

Python · TypeScript · JavaScript · C++ · HTML/CSS

Web & Frontend

React · Three.js · WebGL/WebGPU · Vite · Tailwind CSS · WebSocket

Scientific Computing

NumPy · SciPy · SymPy · Numerical PDE solvers · Monte Carlo methods

Machine Learning

PyTorch · Graph Neural Networks · Diffusion Models · VAEs · Neural ODEs · PINNs · LLM Evaluation (RLHF)

Backend & Infrastructure

FastAPI · REST APIs · Docker · Git · CI/CD · Agentic Coding Tools

Spoken Languages

German (native) · English (C1, full professional proficiency)

SELECTED PROJECTS

Mesh Embeddings & DDG. GPU mesh embedding into hyperbolic and spherical spaces by spring-mass energy minimization, with a discrete differential geometry operator library. Computational basis for the Bridges 2026 paper.

Graph ML Lab. A PyTorch framework for spatial graph generation, built from scratch: GCN, GAT, graph VAE, and joint discrete-continuous diffusion over 3D graph structures. No external GNN libraries. github.com/erik2810/ml-projects

Mesh-Based Physics Simulator. A fully differentiable physics engine in PyTorch that maps mesh topology to particle-spring systems, with energy-based forces and end-to-end backpropagation through the dynamics.

PINN Solver. A Physics-Informed Neural Network solver for 10 PDEs (Burgers, Heat, Wave, KdV, and others) with a PyTorch training backend and client-side JavaScript inference. github.com/erik2810/pde-solver

DiffQFT. Differentiable quantum field theory in AdS_2 , with neural surrogates that replace Monte Carlo integration, a PINN solver, and a FastAPI backend. github.com/erik2810/DiffQFT

Knitted Models. A computational geometry engine that generates woven strand patterns on quad meshes using half-edge data structures and spline interpolation, rendered in Three.js with GPU path tracing.

Neural ODE Engine. A network that learns chaotic Lorenz dynamics in real time, with backpropagation, an RK4 integrator, and an Adam optimizer written by hand in JavaScript. github.com/erik2810/differentiable-physics-engine

Full portfolio with live demos at erik2810.github.io/projects

REFERENCES

Available upon request.